

Advanced Algorithms (Quiz 4)

Course Code: CSE426

Total Marks: 20

Section 2

ID: _____ Name: _____ Serial: _____

Questions

- Q1. (10 Marks)** Construct a unique tree from the following sequence: 4,6,4,5,6,6,7,7. You need to draw the tree and write the list of marked vertices.
- Q2. (10 Marks)** Find the size of the maximum independent set from the tree that you constructed in Question 1. Consider 5 as the root vertex. You need to write the maximum independent set size on each vertex. Also find the set.
- Q3. (2 Marks)** What could be the maximum size of independent set possible on a 8 vertex tree? Draw an example tree that supports your answer.
- Q4. (3 Marks)** Let's take a hypothetical scenario. You find out that each n length string can be turned into exactly two different labeled trees, where n is the number of vertices. Assuming other conditions to be the same, what is the number of distinct labeled tree possible in this scenario? You need to show necessary calculations.